EVOLUTIONARY COMPUTATION

Lecture Notes_3

Fitness Landscapes and Local Optima

Topics Covered
Fitness Landscapes: Configuration spaces, Properties of landscapes
Concepts of Local Optima, Basins of Attraction

1. Fitness Landscapes

Fitness landscapes are representations of how fitness values change across different configurations or solutions in an optimization problem. These are especially important in fields like evolutionary computation, machine learning, and physics.

1.1 Configuration Spaces

- **Definition**: A configuration space represents all possible states or solutions of a system.
- **Examples**:
  - In optimization: All possible values of variables in a function.
  - In physics: All possible arrangements of particles in a system.

1.2 Properties of Landscapes

1. **Fitness Value**: A numerical value assigned to each point (configuration) in the space.
2. **Topology**: The geometric arrangement of points and their connections.
3. **Smoothness**: How gradually or abruptly fitness values change across the landscape.
   - Smooth landscapes often have fewer local optima.
   - Rugged landscapes have many peaks and valleys.
4. **Dimensionality**: Higher-dimensional landscapes are more complex and harder to visualize.
5. **Neutrality**: Regions where fitness remains constant across configurations.

1.3 Examples of Fitness Landscapes

- A **hill** represents a fitness peak (optimal solution).
- A **valley** represents lower fitness (suboptimal solutions).
- A **plateau** represents a flat region with no improvement or degradation in fitness.

2. Local Optima and Basins of Attraction

2.1 Concepts of Local Optima

- **Definition**: A solution is a local optimum if no neighboring solution has a higher fitness value.
- **Global Optimum**: The highest fitness value in the entire landscape.
- **Challenges**: Algorithms may get "stuck" at local optima, failing to find the global optimum.

2.2 Basins of Attraction

- **Definition**: A region in the fitness landscape where any starting point eventually leads to the same local optimum under a specific optimization algorithm.
- **Significance**: Basins determine the behavior of optimization methods and the likelihood of finding global optima.
- **Examples**:
  - Wide basins with steep slopes are easier for algorithms to navigate.
  - Narrow basins may require precise tuning of parameters.

Lab Activities

Activity 1: Visualizing Fitness Landscapes

- **Objective**: Understand the structure of fitness landscapes.
- **Steps**:
  1. Write a program to generate 2D or 3D fitness landscapes for mathematical functions like $f(x) = \sin(x) + \cos(y)$.
  2. Plot landscapes using tools like Matplotlib or MATLAB.
- **Expected Outcome**: Visual insights into peaks, valleys, and plateaus.

Activity 2: Simulating Optimization

- **Objective**: Explore how optimization algorithms traverse fitness landscapes.
- **Steps**:
  1. Implement a hill-climbing or simulated annealing algorithm.
  2. Test the algorithm on generated landscapes.
  3. Observe convergence to local or global optima.
- **Tools Needed**: Python, NumPy, Matplotlib.

Extra Activities

1. **Case Study**: Analyze the fitness landscape of a genetic algorithm applied to the traveling salesman problem.
2. **Research Task**: Investigate the role of basins of attraction in neural network optimization.

Use Cases

1. **Machine Learning**: Fitness landscapes guide hyperparameter tuning.

2. **Evolutionary Biology**: Understanding fitness peaks explains species adaptation.
3. **Optimization**: Engineering problems use landscapes to design efficient systems.

Scenario-Based Questions

1. If an algorithm frequently converges to a local optimum, how can the landscape's topology explain this behavior?
2. How do rugged landscapes impact the performance of genetic algorithms?

Discussion Questions

1. How does dimensionality affect the complexity of fitness landscapes?
2. What strategies can help overcome the challenge of local optima in optimization problems?

Case Analysis Questions

1. Compare the impact of smooth vs. rugged fitness landscapes on optimization techniques like simulated annealing.
2. Analyze the basin of attraction's role in guiding a particle swarm optimization algorithm.

Practice Questions

Conceptual

1. Define a fitness landscape and explain its components.
2. What is the difference between a local and a global optimum?

Mathematical

1. For $f(x, y) = -x^2 - y^2$, identify the global optimum and describe the fitness landscape.

2. Sketch the fitness landscape of $f(x) = \sin(x) + x^2$. **Algorithmic**

1. Write pseudocode for a hill-climbing algorithm navigating a 1D fitness landscape.
2. Implement simulated annealing to avoid getting stuck in local optima.

Sentence Questions

1. Define a basin of attraction in the context of fitness landscapes.
2. Why is smoothness an important property of a fitness landscape?

3. Explain why higher-dimensional landscapes are more challenging for optimization.

Solved Problems on Fitness Landscapes, Local Optima, and Basins of Attraction

1. Fitness Landscapes: Configuration Spaces and Properties

Problem 1: Understanding Configuration Spaces

A simple fitness function is given by $f(x) = -x^2 + 4x + 6$, where x represents possible configurations in the solution space.

- **Task**:
  1. Identify the configuration space for $x \in [-2, 5]$.
  2. Plot the fitness landscape.
  3. Identify the local and global optima.

Solution:

1. **Configuration Space**:
   - $x \in [-2, 5]$: All possible values of x within the interval define the configuration space.
2. **Fitness Landscape**:
   - Rewrite the function:
     $f(x) = -x^2 + 4x + 6$.
   - Derive the vertex to find the critical point using $x = -\frac{b}{2a}$:
     $a = -1, b = 4$.
     $x = -\frac{4}{2(-1)} = 2$.

3. **Fitness Value at the Critical Point**:
   - Substitute $x = 2$ into $f(x)$:
     $f(2) = -(2)^2 + 4(2) + 6 = -4 + 8 + 6 = 10.$
4. **Check the Interval Endpoints**:
   - $f(-2) = -(-2)^2 + 4(-2) + 6 = -4 - 8 + 6 = -6.$
   - $f(5) = -(5)^2 + 4(5) + 6 = -25 + 20 + 6 = 1.$
5. **Plot the Landscape**:
   - The function is a parabola opening downward. The global maximum (global optimum) is at $x = 2$ with $f(2) = 10$.

Answer:

- **Global Optimum**: $x = 2$, $f(2) = 10$.
- **No Local Optimum other than the global optimum** since the function is a smooth parabola.

2. Local Optima and Basins of Attraction

Problem 2: Exploring Local Optima

A fitness function for a rugged landscape is given as $f(x) = \sin(x) + 0.1x$ for $x \in [0, 10]$.

- **Task**:
  1. Identify local optima in the interval.
  2. Approximate basins of attraction for each local optimum.

Solution:

1. **Identify Local Optima**:
   - The derivative of $f(x)$:
     $f'(x) = \cos(x) + 0.1.$
   - Set $f'(x) = 0$:
     $\cos(x) = -0.1.$
     Solving $\cos(x) = -0.1$ within $[0, 10]$ gives:
     $x_1 \approx 1.67, x_2 \approx 4.61, x_3 \approx 7.60, x_4 \approx 10.57.$

2. **Classify Local Optima**:
   - Compute the second derivative $f''(x) = -\sin(x)$.
   - At $x_1 = 1.67$, $f''(1.67) = -\sin(1.67) \approx -0.99 < 0$, so it's a local maximum.
   - At $x_2 = 4.61$, $f''(4.61) = -\sin(4.61) \approx 0.99 > 0$, so it's a local minimum.

   Repeat for other critical points.

3. **Basins of Attraction**:
   - Each critical point has an interval around it where the fitness value converges to that point.
   - For $x_1 = 1.67$, the basin might be $[0, 3.14]$, assuming symmetry and slope direction.

Answer:

- **Local Maxima**: $x = 1.67, 7.60$.
- **Local Minima**: $x = 4.61$.
- **Basins of Attraction**:
  - Basin for $x = 1.67$: $[0, 3.14]$.
  - Basin for $x = 7.60$: $[6.28, 10]$.

3. Basins of Attraction Visualization

Problem 3: Basins of Attraction for Optimization Algorithms

An algorithm searches for the global maximum of $f(x) = -x^4 + 4x^3 - 2x^2 + x$ over $x \in [-2, 3]$.

- **Task**:
  1. Find critical points and classify them.
  2. Identify basins of attraction for local optima.

Solution:

1. **Find Critical Points**:
   - $f'(x) = -4x^3 + 12x^2 - 4x + 1$.
   - Solving $f'(x) = 0$: Critical points are approximately $x = -0.34, 1, 2.34$.
2. **Classify Critical Points**:
   - Compute $f''(x) = -12x^2 + 24x - 4$.
   - At $x = -0.34$, $f''(-0.34) = -12(-0.34)^2 + 24(-0.34) - 4 \approx 6.92$, indicating a local minimum.
   - At $x = 1$, $f''(1) = -12(1)^2 + 24(1) - 4 = 8$, indicating a local maximum.
3. **Basins of Attraction**:
   - Each local optimum has a basin defined by the points where the slope leads to that optimum.

Answer:

- **Local Maximum**: $x = 1$.
- **Local Minimum**: $x = -0.34, 2.34$.
- **Basins**:
  - Basin for $x = 1$: Approx. $[0.5, 1.5]$.